FOR APPROXIMATELY THREE CENTURIES, from the early 1500s through to the 1830s, naval warfare was dominated by three-masted ocean-going sailing ships, armed with cannon. Although the tactics of grappling and boarding continued to hold their place in war at sea, from the 17th century the dominant mode of combat became the exchange of broadsides—volleys of cannon fire designed to bring down masts and rigging, hole the hull, and kill or wound the crew of an enemy vessel. European navies fought battles in the East Indies, the Caribbean, and the mid-Atlantic, as well as the Mediterranean and north and west European coastal waters. Sea power grew from an adjunct to land operations into a major source of imperial dominance in its own right.



The Golden Hind
The *Golden Hind* was the Elizabethan galleon Sir Francis Drake used to circumnavigate the globe (1577–80) and launch numerous raids on the Spanish. It carried an armament of 22 guns and had a top speed of 8 knots (15kph). This full-size replica was launched in 1973.

FIGHTING MACHINES

The design of warships and combat tactics evolved subtly rather than dramatically through the age of sail. During the course of the 16th century ungainly carracks, conceived as large floating fortresses, were supplanted by sleeker galleons. The need to bring guns to bear in broadsides led to the adoption of the line of battle, and by the late 17th century the dominant warships were "ships of the line", graded into first-, second-, and third-rate by their weight of cannon. These were extraordinary fighting machines. Even in the 16th century there were "great ships"—large vessels built for prestige as much as combat—carrying more than 50 guns, but by the late 18th century a three-deck first-rate ship of the line would carry from 100 to 130 guns. Such ships needed large crews packed into their decks because of the number of hands required to sail the vessel and to man the cannon in battle. In order to stay at sea for long periods they also had to carry substantial supplies of all essentials.

MANNING SHIPS

Fleets of sail were extremely expensive to build, maintain, and supply. Any country that wanted to have a dominant navy needed an extensive organization of dockyards for repair and shipbuilding, an administrative system to supply munitions and food, and a pool of sailors to draw on, since only experienced seamen were really useful on board. No country could afford to maintain a full navy in peace and war. Until the mid-17th century, merchant ships, under their peacetime captains, were pressed into frontline service when war broke out. This practice was discontinued because of the superior performance of purpose-built warships





